DeepSEM-GLA v1: Study Protocol and Pre-registration

Version 1.1 · 10 October 2026 · **locked as pre-registration**

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How to use this document. This protocol is written so that the whole study can be executed once, in the order of Section 10, without returning to design decisions. Every number has a source and an evidence grade (Section 2). Anything not fixed here is decided by a rule written here. Any departure is recorded in a deviations log (Section 11) with its date and reason, and reported in the paper.

Contents

1 Objective and hypotheses **2**

2 Evidence grades **2**

3 Design overview **2**

4 Language models **2**
4.1 Inclusion criteria . . . 2
4.2 Included models . . . 3

5 Data: GapQA v1 **3**
5.1 Admission rule . . . 3
5.2 Sources and split mapping . . . 3
5.3 Leakage control . . . 4
5.4 Construction parameters . . . 4

6 Model and training **5**
6.1 Prompting . . . 5

7 Evaluation **6**
7.1 Scores, answers and errors . . . 6
7.2 Metrics . . . 6
7.3 Statistics . . . 6

8 Decision rules (fixed before any training run) **7**

9 Runs and compute budget (150 GPU-hours) **7**

10 Execution phases **7**
10.1 Interruptions (pausing the computer) . . . 8

11 Documentation and reproducibility **8**

12 Implementation work packages **9**

13 Known risks **9**

14 Changes from version 1.0 **9**

1 Objective and hypotheses

DeepSEM-GLA asks whether a language model answers and abstains better when the size and location of omitted context are represented in the architecture (a magnitude-bearing gap token at true positions) rather than in the text (a literal "[g tokens omitted]" marker) or not at all.

H1 (confirmatory; gate, one model).
On Qwen2.5-1.5B-Instruct and the pooled in-domain in-range test set, the gap-token arm G has lower AURC than the literal-marker arm B1, beyond what true positions alone (B1P) provide. Decided by the rules of Section 8.

H2 (confirmatory; generality).
The difference AURC(B1)−AURC(G) on the pooled in-domain in-range test set is positive across model families and sizes (six models, three families, 0.5B–3.2B). Decided by the rule in Section 8.

H3 (secondary, pre-specified; transfer).
On each of the four held-out sets, the primary model's G arm has a higher AUROC$(s; y_{\text{rec}})$ than its B1 arm (seed-averaged), tested with a cluster bootstrap and Holm correction over the four sets.

H4 (exploratory; coverage readout).
The training-free coverage statistic of Section 7 separates recoverable from unrecoverable items. Reported descriptively against a pre-stated reference value.

H1 and H2 carry the claims of the paper. H3 and H4 are reported in full whatever their outcome, and Stage 2 and the held-out evaluation run under every outcome of H1, so the study runs once.

2 Evidence grades

A: supported by peer-reviewed work in a comparable setting. **B**: supported by analogy, a general principle, or a widely used but non-peer-reviewed convention. **C**: our design choice; acceptable only when fixed before results are seen and its influence is reported (stratified results or ablation).

3 Design overview

	Stage 1: gate	Stage 2: generality	Held-out evaluation
Question	H1, H4	H2	H3
Models	Qwen2.5-1.5B-Instruct	5 further models (+1 conditional)	all trained checkpoints and zero-shot anchors
Arms	B0, B1, B2, B1P, G0, G	B1, G	as trained
Seeds	B1 and G: 0, 1, 2; B1P: 0, 1; B0, B2, G0: 0	0	—
Runs	11	10 (+2 conditional)	—
Data	GapQA v1 train/dev/test	same	4 held-out sets

The six arms render one stored item in six ways: B0 nothing at the gap, renumbered positions; B1 literal marker, renumbered; B2 nothing, true positions; B1P literal marker placed inside the gap, true positions; G0 gap token without magnitude, true positions; G gap token with magnitude, true positions.

What a seed controls. Seed i sets the LoRA initialisation, the LoRA dropout masks, the noise in the initial gap embedding, and the order of training items in each epoch. Arms with the same seed index share all of these, so the comparison G_i versus B1_i is paired; only the rendering and the gap module differ [14].

4 Language models

4.1 Inclusion criteria

A model is included if it (i) has open weights and an instruction-tuned release; (ii) has at most 3.2B parameters, so it trains with LoRA on one 12 GB GPU; (iii) has a trained context of at least 32k tokens, because true positions in GapQA v1 reach about 15.3k tokens (in the extrapolation slice: 3000 observed tokens plus up to three gaps of up to 4096 tokens) and positions beyond the trained range would confound the comparison; (iv) uses the projection names `q,k,v,o,gate,up,down`, so one LoRA configuration applies to all; and (v) has an unused special token that can serve as the gap placeholder without resizing the vocabulary.

4.2 Included models

Model	Params	Context	Licence (gated)	Placeholder	Role	Notes
Qwen2.5-1.5B-Instruct [59]	1.54B	32k	Apache-2.0 (no)	`<|fim_pad|>`	Stage 1 primary	tied embeddings
Qwen2.5-0.5B-Instruct [59]	0.49B	32k	Apache-2.0 (no)	`<|fim_pad|>`	Stage 2	size axis
Qwen2.5-3B-Instruct [59]	3.09B	32k	Qwen Research (no)	`<|fim_pad|>`	Stage 2	size axis; derived weights research-only
Llama-3.2-1B-Instruct [48]	1.23B	128k	Llama 3.2 Community (yes)	`<|reserved_special_token_0|>`	Stage 2	family 2
Llama-3.2-3B-Instruct [48]	3.21B	128k	Llama 3.2 Community (yes)	`<|reserved_special_token_0|>`	Stage 2	family 2
Gemma-3-1B-it [24]	1.0B	32k	Gemma (yes)	`<unused0>`	Stage 2	family 3; 5:1 local/global attention; local windows count slots, not positions
Granite-3.1-2B-instruct [26]	2.5B	128k	Apache-2.0 (no)	`<fim_pad>`	conditional	family 4; run only if budget allows

Excluded (criterion iii): SmolLM2-1.7B-Instruct and Gemma-2-2B-it (8k context). Qwen2.5-7B and Llama-3.1-8B exceed criterion (ii) and the compute budget. **Access actions**: request access to the Llama 3.2 and Gemma 3 repositories on Hugging Face before Phase P0 ends. **Language coverage**: Qwen2.5 lists Arabic but not Indonesian; Llama 3.2 officially supports neither; Granite lists Arabic. Held-out results in Indonesian and Arabic are therefore reported per model and interpreted with this in mind.

5 Data: GapQA v1

5.1 Admission rule

A source is admitted only if the answer-bearing evidence is annotated (span, sentence, paragraph or dialogue turn), so that recoverability y_{rec} is exact by construction. This rule excludes DROP [16] (no evidence annotation) and TriviaQA [38] (distantly supervised evidence). GapQA v0 (S1 + SQuAD) is retained as a pilot and is not mixed with v1.

5.2 Sources and split mapping

Source	Role	Evidence unit	Licence	GapQA split ← official split	Train	Dev	Test
S1 synthetic key–value [47, 34]	in-domain	the key–value sentence	ours	generated; disjoint key vocabularies per split	600	100	350+100
SQuAD v1.1 [60]	in-domain	sentence containing the answer span	CC BY-SA 4.0	train, dev ← train; test ← dev	680	100	350+100
HotpotQA, distractor [68]	in-domain	all supporting-fact sentences	CC BY-SA 4.0	train, dev ← train; test ← dev	680	100	350+100
Natural Questions [43]	in-domain	long-answer paragraph containing a short answer	CC BY-SA 3.0	train, dev ← first 10 train shards of the official release; test ← official dev	680	100	350+100
MuSiQue, answerable [66]	in-domain	all supporting paragraphs	CC BY 4.0	train, dev ← train; test ← dev	680	100	350+100
2WikiMultiHopQA [32]	in-domain	all supporting-fact sentences	Apache-2.0	train, dev ← train; test ← dev	680	100	350+100
QASPER [10]	held-out	evidence paragraphs (text only)	CC BY 4.0	test ← test	—	—	500
TyDi QA GoldP, Indonesian [9]	held-out	sentence containing the answer span	Apache-2.0	test ← dev; distractors ← Indonesian train passages	—	—	$\leq$500
TyDi QA GoldP, Arabic [9]	held-out	sentence containing the answer span	Apache-2.0	test ← dev; distractors ← Arabic train passages	—	—	500
LoCoMo [51]	held-out	evidence dialogue turns; session time stamps	CC BY-NC 4.0	all 10 conversations; non-adversarial questions	—	—	500

Official test splits without public labels are never used. In-domain test counts are in-range plus extrapolation items. Totals: train 4000, dev 600, in-domain test 2100 in-range and 600 extrapolation, held-out about 2000. TyDi Indonesian has 565 dev questions, so it contributes all eligible items up to 500. LoCoMo time stamps are stored for version 2 (duration input τ) and unused in v1. QASPER uses answerable questions with text evidence only.

5.3 Leakage control

- **Wikipedia titles.** Let T be the set of Wikipedia titles appearing anywhere in a test context (question paragraphs, distractors and padding). No train or dev context may contain a title in T; training items that would are removed, and distractor pools for train and dev are drawn from titles outside T. Train and dev are separated by a fixed hash of titles, as in v0. Counts of removed items are logged.
- **Closed-book filter.** Run *first*, on the candidate pool of every source, in-domain and held-out: a question is removed if *any* of the seven listed models, Granite included, answers it without context (EM over gold answers and aliases) [46]. Pairs and class balance are formed afterwards, so target counts are met from the filtered pool.
- **Lexical floor.** Recoverability can partly be read from question–context word overlap. A training-free baseline (BM25 score of the best observed sentence [62]) is scored on every test set; its AURC and AUROC are reported as the floor that any arm must beat.

5.4 Construction parameters

Parameter	Value	Grade	Source and safeguard
Storage	text with character offsets	C	Model-agnostic, so all models see the same deletions. Lengths are sampled in a reference tokenizer (Qwen2.5) and re-counted in each model's tokens at render time; the B1 marker and the G magnitude use the model's own count. KAT checks round-trips.
Observed length T_{obs}	3000 reference tokens	C	Fits one 12 GB GPU at batch 1; confirmed by benchmark. Not a long-context claim.
Number of gaps K	$\min(1 + \text{Poisson}(2), 5)$	C	Mean about three gaps; the cap keeps true positions below 13.3k in-range. Results stratified by K.
Gap length g	LogUniform(16, 2048)	B	Log-scale sampling for scale parameters [5]; 16 keeps the literal marker shorter than the hole. Stratified by g bin.
Extrapolation slice	$g \in (2048, 4096]$, $K \leq 3$	A	Train-short/test-long [58, 2]. This is per-gap *magnitude* extrapolation (positions stay below 15.3k); it is compared with in-range items restricted to $K \leq 3$.
Composition	compose-to-fit	B	Natural long documents (NQ, QASPER, LoCoMo) use a contiguous window around the evidence; short ones are padded with distractor documents from the same source and split [34].
Evidence in window	all evidence units must fit in the window	C	Items whose evidence cannot fit (long-range LoCoMo or QASPER questions) are excluded; exclusion rates are reported per source.
Multi-evidence items	$y_{\text{rec}} = 1$ iff every evidence unit survives	C	HotpotQA, MuSiQue, 2Wiki, QASPER, LoCoMo; a deleted evidence unit lies entirely inside one gap.
Pairing	length-matched pairs within one source	A	Same gap multiset for one $y_{\text{rec}} = 0$ and one $y_{\text{rec}} = 1$ item [23, 28, 19]; checked exactly per pair.
Class balance	y_{rec} 50/50 per source	B	Fixed prevalence keeps AURC comparable [22].
Sentence splitting	abbreviation-aware; Arabic and Indonesian punctuation rules	C	KAT: answer inside its evidence for every item.
Training target	first gold answer in the official order	C	NQ: first annotator's first short answer; aliases used only for scoring.
Data seed	20261009	—	Recorded with SHA-256 of every file and a datasheet [20].

6 Model and training

All models use one configuration. Within-model comparisons are therefore exact; cross-model comparisons use within-model differences only (Section 7.3).

Parameter	Value	Grade	Source
Gap embedding	$e_{\text{gap}}(g) = e_{[\text{GAP}]} + W_g\phi(u)$, $u = \log(1+g)$	—	Design under test. Injected through `inputs_embeds` as separate parameters; the tied placeholder row is never modified.
Features $\phi(u)$	$[u/u_{\max}; \sin(2^k u); \cos(2^k u)]_{k=0}^{7}$, 17 dims, $u_{\max} = \log(1+2048)$	—	Fourier features [65, 52]; the linear term lets magnitude extrapolate, because periodic features alias outside the training range.
W_g init	zero	A	[3, 35]; G starts identical to G0.
$e_{[\text{GAP}]}$ init	mean embedding + noise (std 0.02× embedding std)	B	[31] (technical note).
Placeholder at decoding	logit set to $-\infty$	—	The placeholder can never be generated.
Decision score	$s = p_A/(p_A + p_B)$ at the first assistant position	A	p_A, p_B: probabilities of the first tokens of `ANSWER:` and `ABSTAIN`, verified distinct per tokenizer [39, 37].
LoRA targets	all 7 projections	A	[13, 6].
Rank, scale	$r = 16$, $\alpha = 32$	A	[13, 6] ($\alpha = 2r$).
LoRA dropout	0.05	A	[13].
LR (LoRA / gap)	2×10^{-4} / 10^{-3}	A / B	[13, 6] / [45].
Optimizer	AdamW, $\beta = (0.9, 0.999)$, wd 0	A	[50].
Schedule	3% warm-up, cosine	B	[49, 25, 1].
Batch / epochs	1×8 accumulation / 2	C	Identical across arms and models; training curves reported.
Gradient clipping	1.0	B	[56]; grad norm logged.
Precision, memory	bf16, gradient checkpointing	A	[40, 8].
Out-of-memory fallback	4-bit QLoRA for that model only	A	[13]; recorded as a deviation.
Checkpoint selection	none; final checkpoint	A	Avoids selection on evaluation data [14]; dev is monitored only.
Attention mask	explicit all-ones mask in every call	—	Prevents packed-sequence inference across position jumps (KAT-3d, per model).
Failed run	NaN/inf loss, or final training loss not below the first logged loss	C	Restart once from scratch with the same seed; a second failure is recorded and the affected contrasts are reported as missing.
Determinism	seeds fixed; CUDA kernels not forced deterministic	B	Residual non-determinism is acknowledged [57]; model and dataset revisions are pinned by commit hash at download.

6.1 Prompting

One system text is used for all models and arms: *"Answer the question using only the context. Parts of the context may be missing. If the answer is not in the context, reply ABSTAIN. Otherwise reply ANSWER: followed by the answer."* The prompt is assembled with each tokenizer's own chat template, using a sentinel string that is split to insert the context at token level. Templates without a system role (Gemma 3) place the system text at the start of the first user turn, as the template does. The Llama 3.2 template's date field is fixed to "9 Oct 2026". The full rendered prompt of one item per model is stored with its hash. Instructions are in English for every language; context and question keep their original language. Zero-shot anchors use the same prompt.

7 Evaluation

7.1 Scores, answers and errors

For each test item the model returns s in one forward pass. For in-domain items an answer string is obtained by appending the tokens of `ANSWER:` to the prompt and decoding greedily for at most 24 tokens (stopping at the end-of-turn token or a newline). The budget of 24 tokens is checked against the 99th percentile of gold answer lengths in every tokenizer (grade C). An answered in-domain item is an error $e = 1$ if $y_{\text{rec}} = 0$, or if $y_{\text{rec}} = 1$ and the best exact match over all gold answers and aliases, after SQuAD normalisation [60], is zero.

7.2 Metrics

Metric	Use	Source
AURC with EM-based errors	primary, in-domain (H1, H2)	[22]
AUROC$(s; y_{\text{rec}})$; AURC with $e = 1 - y_{\text{rec}}$	primary, held-out (H3), answer-free; also reported in-domain	[29, 41]; separates abstention from answer quality, which differs across languages and answer styles
Selective risk at 80% and 90% coverage	secondary	[18, 21]
Hallucination rate on $y_{\text{rec}} = 0$ at $s \geq 0.5$; EM and F1 on $y_{\text{rec}} = 1$	secondary	[60]
ECE (15 bins) of s against $1 - e$	secondary	[27, 54]
Lexical floor (BM25)	reference for every test set	[62]
Coverage readout $\bar{m}$	H4: mean attention mass, over all layers and heads, from the first assistant position onto gap or marker keys; AUROC$(1 - \bar{m}; y_{\text{rec}})$; arms B1, B1P, G0, G, seed 0, primary model	[36, 67]; reference value 0.70 (grade C), reported with its 95% interval

All metrics are reported per source, per slice (in-range, extrapolation), and stratified by K and by g bin. Zero-shot anchors (B0 and B1 renderings, no training) are scored for every model.

7.3 Statistics

- **Resampling units.** In-domain: length-matched pairs. QASPER: papers. LoCoMo: conversations (a cluster bootstrap; with ten clusters its intervals are reported as approximate). TyDi: items. [17, 11].
- **Seeds.** For an arm with several seeds, AURC is computed per seed and averaged; inside each bootstrap replicate the same resampled items are used for every seed and arm [14, 61].
- **Intervals and p-values.** Percentile intervals from 1000 replicates; two-sided p-values $p = 2\min\{P^*(\Delta \leq 0), P^*(\Delta \geq 0)\}$ from 10 000 replicates [17, 42, 15].
- **Equivalence.** TOST via the 90% interval, margin $\delta = 0.01$ AURC [63, 44].
- **Multiplicity.** Holm correction over the gate family $\{\text{G}-\text{B1}, \text{G}-\text{B1P}, \text{G}-\text{G0}, \text{B1P}-\text{B1}\}$, and separately over the four held-out sets in H3 [33]. If an arm is dropped (Section 9), its contrasts leave the family.
- **Across models (H2).** Per-model differences AURC(B1)$-$AURC(G) with bootstrap variances are pooled by a random-effects model with the Hartung–Knapp–Sidik–Jonkman interval, which keeps nominal coverage with few studies [12, 30, 64]. Family-level pooled estimates and every per-model estimate are shown. Sensitivity analyses: (a) each per-model variance inflated by the Stage-1 seed-to-seed variance of the same difference, because Stage 2 has one seed; (b) leave Gemma out (its 5:1 local/global attention windows count sequence slots, not position ids); (c) leave one family out.
- **Power check (before training).** On zero-shot B0 and B1 of the primary model, the half-width of the 90% paired-bootstrap interval of AURC(B0)$-$AURC(B1) must be at most $\delta/2$, so that TOST has adequate power when the true difference is near zero [7, 44]. Otherwise the in-domain test set is enlarged to 700, then 1050 items per source where the official split allows. If the target is still missed, equivalence-based outcomes are labelled underpowered (Section 8).

8 Decision rules (fixed before any training run)

Let $\Delta\text{AURC}(a) = \text{AURC}(\text{B0}) - \text{AURC}(a)$ on the pooled in-domain in-range test set of the primary model, seed-averaged. Rules are applied in this order; the first that holds is the outcome.

1. **Degenerate.** $\Delta\text{AURC}(\text{G}) \leq 0.02$ (grade C). The task is reported as too easy or too hard; no outcome is called.
2. **Outcome 1, the literal marker suffices.** $\Delta\text{AURC}(\text{B1}) \geq 0.9\,\Delta\text{AURC}(\text{G})$ and B1 and G are TOST-equivalent at $\delta = 0.01$. The ratio 0.9 is grade C, fixed in advance [55].
3. **Outcome 2, positions carry the gain.** $\Delta\text{AURC}(\text{B1P}) \geq 0.9\,\Delta\text{AURC}(\text{G})$ and B1P and G are TOST-equivalent.
4. **Outcome 3, the representation carries the gain.** G$-$B1 and G$-$B1P are both significant after Holm ($p < 0.05$) in favour of G (an intersection–union test [4]), and the per-seed differences $\text{G}_i - \text{B1}_i$ have the same sign for $i = 0, 1, 2$.
5. **Inconclusive.** Otherwise. If the power check failed, an outcome that needed equivalence is reported as "inconclusive, underpowered for equivalence". No seeds or items are added after results are seen.

H2. Supported if the pooled HKSJ 95% interval of AURC(B1)$-$AURC(G) lies above zero *and* at least five of the six per-model point estimates are positive; contradicted if the interval lies below zero; otherwise inconclusive. **H3.** For each held-out set, supported if AUROC(G)$-$AUROC(B1) is positive with Holm-adjusted $p < 0.05$. **H4.** Reported against the reference AUROC of 0.70 with its interval; no decision depends on it. **Blinding.** No Stage-1 test metric is computed until all Stage-1 runs have finished; the gate memo is produced by a script from the logged outputs. Stage 2 and the held-out evaluation run under every outcome.

9 Runs and compute budget (150 GPU-hours)

Planning cost of one Qwen2.5-1.5B training run: 4.5 h (4000 items, 2 epochs, about 3.3k tokens, RTX 3080 Ti; the plan's range was 3–5.5 h). Other models scale by parameter count, a conservative rule because large embedding matrices add little compute when only target logits are computed. Evaluation of one 1.5B checkpoint: about 0.67 h (scoring about 4700 test items plus greedy answers on recoverable in-domain items). Bootstrap and meta-analysis run on the CPU.

Item	Units	Cost rule	GPU-h
Stage 1 training	11 runs	11×4.5	49.5
Stage 1 evaluation and attention diagnostic	11 checkpoints	$11 \times 0.67 + 2$	9.4
Stage 2 training (Qwen 0.5B, Qwen 3B, Llama 1B, Llama 3B, Gemma 1B)	10 runs	$2 \times 5.86 \times 4.5$	52.7
Stage 2 evaluation	10 checkpoints	$2 \times 5.86 \times 0.67$	7.9
Zero-shot anchors (6 models, B0 and B1, scoring only)	12	$2 \times 6.86 \times 0.26$	3.6
Closed-book filter, union over 7 models	≈18k questions	8.48×0.5	4.2
KATs, memory benchmarks, smoke runs	7 models	—	3.0
Total planned			130.3
Contingency (about 15% of the planned total)			19.7

Parameter-count factors relative to 1.54B: Qwen 0.5B 0.32, Qwen 3B 2.01, Llama 1B 0.80, Llama 3B 2.08, Gemma 1B 0.65, Granite 1.62. Sum over the six core models including the primary one: 6.86; with Granite: 8.48. Granite training and evaluation (about 17 h) runs only if the budget checkpoint shows that much slack. If the power check enlarges the test set, evaluation cost grows proportionally (at most about $2\times$, covered by the contingency).

Budget checkpoint. After Phase P2 the measured time per micro-step replaces the planning value. If the projected total exceeds 150 h, runs are dropped in this pre-registered order until it fits: (1) Granite (already conditional); (2) B1P seed 1; (3) B2 seed 0; (4) G0 seed 0; (5) Qwen2.5-3B in Stage 2. Never dropped: Stage 1 B0 seed 0, B1 seeds 0–2, G seeds 0–2 and B1P seed 0; Stage 2 Qwen2.5-0.5B, Llama-3.2-1B, Llama-3.2-3B and Gemma-3-1B. Dropped arms leave the Holm family; Outcome 2 then rests on B1P seed 0 alone, which is reported.

10 Execution phases

Each phase has an exit criterion; the next phase starts only when it is met.

Phase	Work	Exit criterion
P0	Environment; access to Llama 3.2 and Gemma 3; download 7 models (about 26 GB) and 8 datasets (Natural Questions: 10 train shards and the dev split of the official Hugging Face release; MuSiQue and 2WikiMultiHopQA from the authors' links); pin revisions	All files present with commit hashes and SHA-256; licences recorded.
P1	Union closed-book filter; build GapQA v1; leakage control; freeze	All data KATs pass, including exact gap-multiset equality within every pair and zero test titles in train/dev; datasheet written; exclusion counts logged.
P2	Per model: tokenizer, chat-template and placeholder KATs, decision-token check, KAT-2/3/3d, memory benchmark; smoke runs (B1, G) on the primary model	All KATs pass for every model. Smoke runs are judged per arm only (training loss falls; AUROC$(s; y_{\text{rec}})$ on trained items > 0.9); arms are not compared. A failure is fixed in code only and rerun; a needed hyperparameter change is a logged deviation made before P4. Budget checkpoint applied.
P3	Zero-shot anchors for all models; lexical floor; power check	Power criterion met, or test set enlarged and re-frozen, or the underpowered label recorded.
P4	Stage 1 queue (11 runs); evaluation after the last run; gate memo by script	Outcome stated by Section 8.
P5	Stage 2 queue (10 runs) and evaluation	Per-model estimates, HKSJ pooling and sensitivity analyses computed.
P6	Held-out evaluation of all checkpoints; H3, H4; stratified tables; figures	All tables of the paper generated by scripts.
P7	Write-up; release code, data (licence permitting), adapters	Reproducibility checklist complete.

10.1 Interruptions (pausing the computer)

Stopping the computer is expected and is not a deviation, provided work resumes from a saved state.

- **Training.** Checkpoints are written atomically (to a temporary folder, marked complete, then swapped in) every 100 optimizer steps, about 15–20 minutes; a half-written checkpoint is never used. A planned pause is made by creating the file `runs/STOP` or pressing Ctrl+C: the run saves its state at the last completed optimizer step (model, optimizer, scheduler, data position, random-number states) within seconds, and the queue halts. Rerunning the same command resumes; a test confirms that an interrupted and resumed run ends with the same weights as an uninterrupted one on CPU. A power cut loses at most 100 optimizer steps.
- **Evaluation, closed-book filter and data building.** Outputs are appended per item to files keyed by item id; a restart skips items already done, so no result depends on whether a pause occurred.
- **Recording.** Each pause and resume (time, step) is written to the run log and the run manifest. On GPU, kernels are not forced deterministic, so a resumed run may differ from an uninterrupted one in the last digits; this is the same order of variation as rerunning the same seed and is reported, not corrected.
- **Practical rule.** Disable sleep and automatic updates while a run is active; prefer pausing between runs (the queue's natural boundary), which loses nothing.

11 Documentation and reproducibility

- **Run manifest** per run: code commit, configuration hash, data SHA-256, model repository and revision, library versions (lock file), GPU, seed, wall time, peak memory.
- **Deviations log**: date, phase, what changed, why, and its effect; reported in the paper.
- **Datasheet** for GapQA v1 [20] and **model cards** for released adapters [53].
- **Reproducibility checklist** [57]: every table and figure generated by a script from logged outputs; no hand-entered numbers.
- **Licences**: LoCoMo is non-commercial and Qwen2.5-3B is research-only, so derived artefacts from them are

released under the same terms or not at all.

12 Implementation work packages

WP1 GapQA v1: character-space storage; adapters for SQuAD, HotpotQA, NQ (official release, 10 train shards and dev), MuSiQue, 2Wiki, QASPER, TyDi (id, ar), LoCoMo; title-level leakage control; evidence-in-window rule; multi-evidence recoverability; language-aware splitter; union closed-book filter; lexical floor.
WP2 Multi-model rendering: chat templates via the tokenizer with a sentinel split, fixed Llama date, system text for templates without a system role, per-model placeholder and decision tokens, per-model true positions and gap counts.
WP3 Gap module update: 17-dimensional features with the linear term; placeholder masked at decoding.
WP4 Evaluation harness (resumable per item): scoring, forced-prefix greedy decoding, EM/F1 over aliases, all metrics, pair and cluster bootstrap, TOST, Holm, intersection–union test, HKSJ pooling with sensitivity analyses, stratified tables, attention diagnostic.
WP5 Run manifest, deviations log, a queue covering both stages with the budget checkpoint, failed-run handling.
WP6 Per-model KATs (template round trip, placeholder, decision tokens, KAT-2/3/3d, Gemma window check).

Already done and reused: GapQA v0 pipeline and tests, the gap module, the training loop with resume, metrics with TOST and Holm (65 tests passing).

13 Known risks

- Measured cost above plan: handled by the budget checkpoint and drop order.
- Recoverability partly visible from text overlap, making all arms similar: the lexical floor shows how much; the degenerate rule prevents over-interpretation.
- Weak multilingual ability of some models on Indonesian and Arabic: held-out results are interpreted per model; answer-free primary metrics limit the confound.
- LoCoMo has only ten conversations: its intervals are approximate and it never decides a hypothesis alone.
- A 3B model does not fit in 12 GB: 4-bit QLoRA fallback for that model, logged as a deviation.
- The ratio 0.9, the threshold 0.02, the H4 reference 0.70 and the smoke threshold 0.9 are judgments (grade C) and disclosed as such.

14 Changes from version 1.0

Version 1.1 closes gaps found in a final internal review and an independent adversarial review: hypotheses split into confirmatory and exploratory with decision criteria for H2–H4; seed semantics; decision-rule precedence; Outcome 3 as an intersection–union test; bootstrap p-values and resampling units (pairs, papers, conversations); power check sized for TOST; HKSJ pooling with sensitivity analyses; official split mapping, including the Natural Questions dev source (the full official release, of which ten training shards are used, because the simplified release has no dev split); title-level leakage control over every paragraph; filter-then-pair ordering and Granite in the filter union; K capped at 5 and the extrapolation slice redefined as magnitude extrapolation; evidence-in-window rule; a linear term in the gap features; prompting, answer extraction and scoring over aliases; failed-run, blinding and smoke-run rules; interruption handling; corrected drop list.

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